Loan Default Prediction Analysis

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1. Introduction

Loan default prediction has become increasingly important as financial institutions strive to reduce lending risks and make fair, informed decisions. Since every loan carries some level of uncertainty, identifying borrowers who are more likely to default can help minimize financial losses while ensuring that credit is provided to reliable applicants.

This project uses historical loan data to examine the factors that influence loan repayment and default. Through data exploration, preprocessing, and machine learning techniques, patterns within borrower information are analyzed to build a model capable of predicting the likelihood of loan default. Such predictions can support more accurate risk assessment and contribute to better lending decisions.

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2. Methodology

2a. Data Collection

The dataset used in this project was obtained from Kaggle under the title **"Loan Prediction Based on Customer Behavior."** It consists of a single CSV file, **Training_Data.csv**, which served as the only source of data throughout the analysis. No separate test dataset or sample prediction file was used. Instead, the dataset was divided into training and testing subsets, allowing the model to be trained, validated, and evaluated using different portions of the same data. This approach ensures that the final performance metrics are based on unseen data while making full use of the available dataset.

Property	Value
Rows	252,000
Columns	13 (12 features + target)
Target column	Risk_Flag (1 = defaulted, 0 = did not)
Missing values	0 (across all columns)
Duplicate rows	0

Raw columns: ID, Income, Age, Experience, Married/Single, House_Ownership, Car_Ownership, Profession, CITY, STATE, CURRENT_JOB_YRS, CURRENT_HOUSE_YRS, Risk_Flag.

2b. Data Preprocessing

An initial inspection of the dataset showed that it was already well-structured. No missing values or duplicate records were found among the 252,000 observations, so there was no need for data cleaning in terms of handling incomplete or repeated entries. Although the dataset required no cleaning for missing or duplicate values, additional preprocessing was necessary to ensure the data was suitable for model training and evaluation.

One of the main challenges was the imbalance in the target variable. The dataset contained 221,004 non-default cases (87.7%) and 30,996 default cases (12.3%), meaning that non-defaulters outnumbered defaulters by approximately 7 to 1. Training a model on this distribution could lead to a bias toward predicting the majority class. To address this issue, several class-balancing techniques, including resampling and class weighting, were applied and compared during model development.

Another challenge involved the categorical variables CITY, STATE, and Profession. These features contained a large number of unique categories—317 cities, 29 states, and 51 professions. Using one-hot encoding for these variables would have created a very large number of additional features, increasing the complexity of the model without necessarily improving performance. Instead, target encoding was used to represent these variables in a more compact and informative way.

To evaluate the models fairly, the dataset was divided into training (70%), validation (15%), and testing (15%) sets, resulting in 176,400, 37,800, and 37,800 records, respectively. A stratified split was used to ensure that each subset maintained the same proportion of default and non-default cases as the original dataset. Furthermore, preprocessing steps that learn information from the data—such as target encoding, feature scaling, and resampling—were fitted only on the training set before being applied to the validation and test sets. This prevented information from the evaluation data from influencing the training process and ensured that the reported model performance reflected its ability to generalize to unseen data.

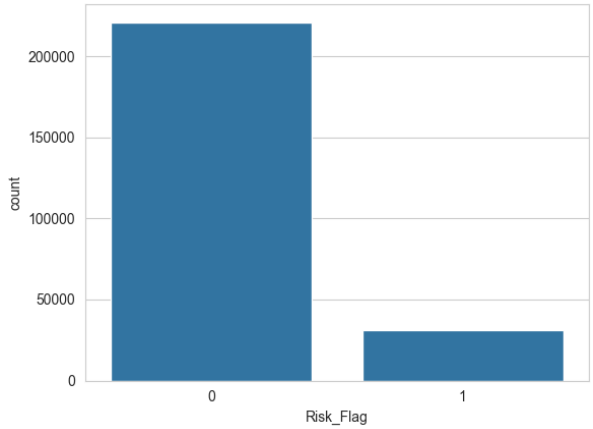


Figure 1. Class distribution of the target variable (Risk_Flag). 221,004 non-defaulters vs. 30,996 defaulters.

2c. Feature Engineering

To improve the model's ability to identify patterns associated with loan default, several new features were created from the existing variables. These engineered features were designed to capture aspects of borrower stability and financial behaviour that may not be evident from the original data alone.

- The **job_stability_ratio** was calculated by dividing the number of years spent in the current job by the borrower's total work experience. This feature reflects the proportion of a person's career spent with their current employer, with higher values indicating greater employment stability.
- Similarly, the **house_stability_ratio** was created by dividing the number of years spent in the current residence by total work experience. This provides an indication of residential stability, which may be associated with lower credit risk.

- An additional feature, **income_per_exp_year**, was calculated by dividing annual income by total years of work experience (plus one to avoid division by zero). This represents a simple measure of earning capacity relative to experience and helps distinguish borrowers with similar incomes but different career lengths.
- The continuous **Age** variable was also grouped into five categories: **under 25**, **25–35**, **35–45**, **45–55**, and **55 years and above**. Converting age into broader groups allows models, particularly tree-based algorithms, to identify meaningful age-related patterns more effectively than using individual age values.
- Categorical variables with relatively few unique values, including **Married/Single**, **House_Ownership**, **Car_Ownership**, and the engineered **age_bucket** feature, were transformed using one-hot encoding. This approach allows the model to use these categorical attributes without introducing an excessive number of additional features.

2d. Exploratory Data Analysis

The exploratory analysis showed that single applicants had a higher loan default rate (**12.53%**) than married applicants (**10.25%**). Although the difference is relatively small, it suggests that marital status may provide useful information when assessing credit risk. This may be because married individuals often have greater financial stability, such as shared household income, making them slightly less likely to default on a loan.

Car and house ownership

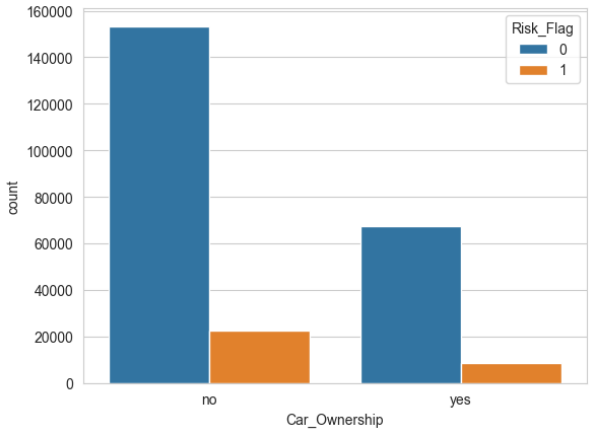


Figure 2. Applicant count by car ownership, split by default outcome.

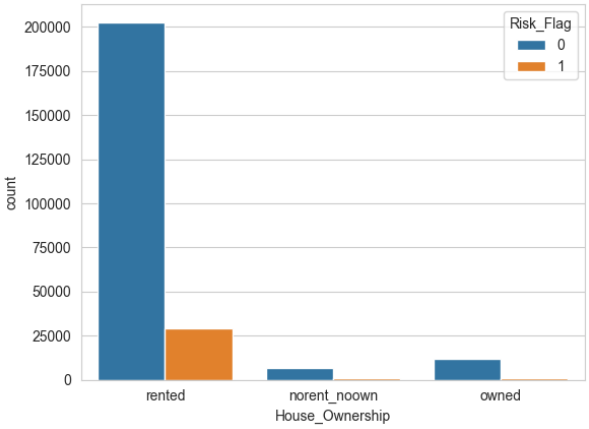


Figure 3. Applicant count by house ownership, split by default outcome.

A chi-square test found a statistically significant relationship between **car ownership** and **loan default** ($\chi^2 = 145.42$, $p < 0.001$). Although the difference in default rates between car owners and non-owners was relatively small, the results suggest that car ownership is associated with loan default and may provide useful information when predicting credit risk.

Numeric feature distributions

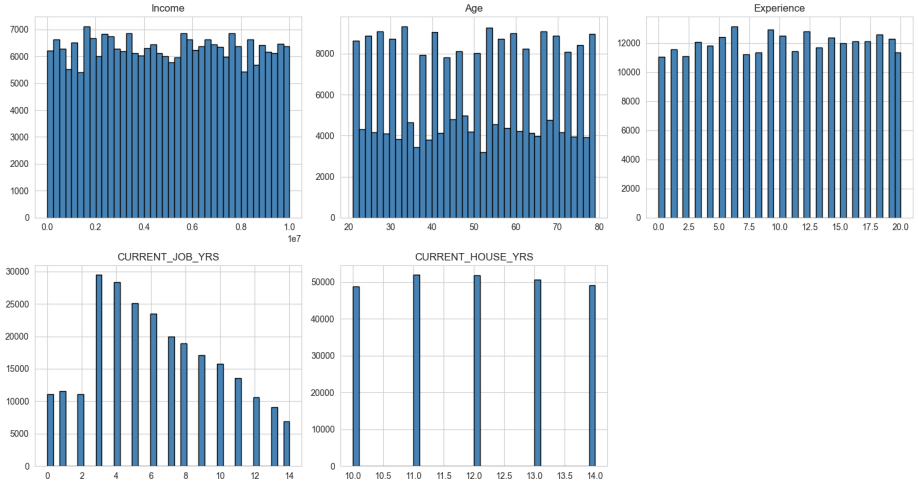


Figure 4. Distributions of Income, Age, Experience, CURRENT_JOB_YRS, and CURRENT_HOUSE_YRS.

The numerical features show different distribution patterns across the dataset. **Income**, **Age**, and **Experience** are fairly evenly distributed, providing a balanced representation of borrowers. However, the uniform distribution of **Income** differs from the pattern typically observed in real-world data, suggesting the dataset may be synthetically generated.

CURRENT_JOB_YRS shows moderate variation, with most applicants having **3–10 years** in their current job. In contrast, **CURRENT_HOUSE_YRS** is concentrated between **10 and 14 years**, indicating limited variation and, therefore, potentially lower predictive value.

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Categorical default rates

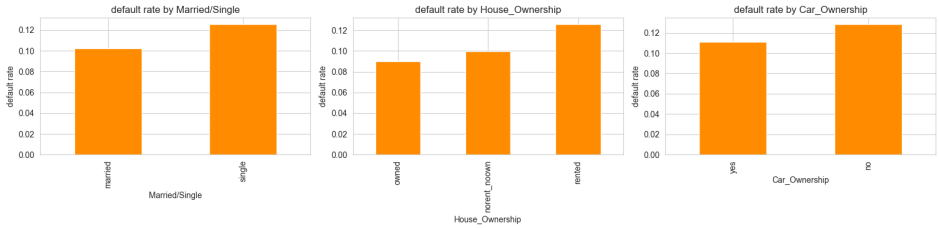


Figure 5. Default rate by Married/Single, House_Ownership, and Car_Ownership.

Profession

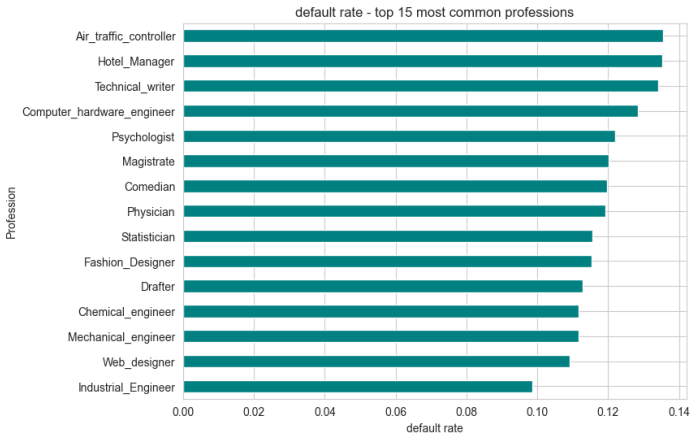


Figure 6. Default rate for the 15 most common professions.

The exploratory analysis showed that loan default rates varied across different professions, including the most common occupations. This suggests that profession contains useful information for predicting loan default.

Correlation structure

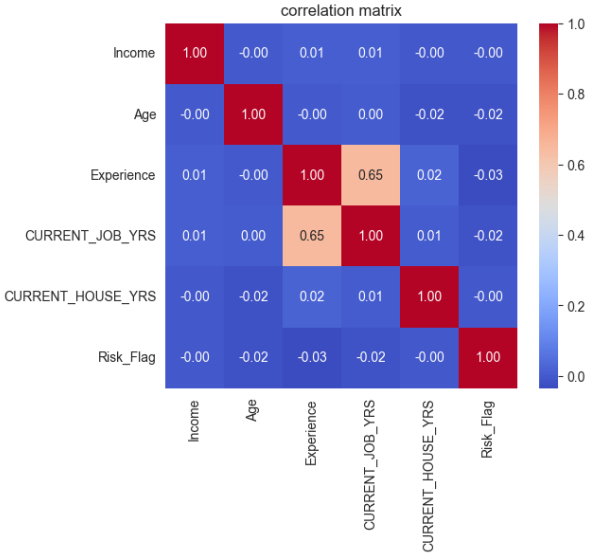


Figure 7. Correlation matrix of the numeric features and the target.

Feature	Pearson r	p-value	Significant?
Income	-0.0031	0.121	No
Age	-0.0218	< 0.001	Yes
Experience	-0.0345	< 0.001	Yes
CURRENT_JOB_YRS	-0.0169	< 0.001	Yes

The correlation analysis showed that the numerical features had only **weak linear relationships** with loan default. **Age**, **Experience**, and **CURRENT_JOB_YRS** were all statistically significant ($p < 0.001$), although their correlation coefficients were small.

2e. Model Development

The objective of this study was to develop a reliable machine learning model capable of predicting whether a borrower is likely to default on a loan. Since the dataset is highly imbalanced—with significantly fewer default cases than non-default cases—special attention was given to handling class imbalance before model training. A model trained on the original data would naturally favor the majority class, leading to poor identification of actual defaulters, which is the primary concern for lenders.

Handling Class Imbalance

The training dataset consisted of **176,400 observations**, with approximately **7.1 non-default loans for every default loan**. Such an imbalance can bias machine learning algorithms toward predicting the majority class. To overcome this issue, several resampling techniques and cost-sensitive learning approaches were evaluated.

The distribution of the classes before and after applying each imbalance-handling technique is shown below.

Each technique addresses the imbalance differently:

- **SMOTE (Synthetic Minority Oversampling Technique)** generates synthetic default cases instead of simply duplicating existing ones, enabling the model to learn more representative decision boundaries.
- **SMOTETomek** combines SMOTE with Tomek Links removal, eliminating ambiguous observations near class boundaries after oversampling. This often improves class separation and reduces noise.
- **Random Undersampling** balances the dataset by randomly removing majority-class observations. Although computationally efficient, valuable information from the majority class may be lost.
- **Class Weighting** does not alter the data distribution. Instead, higher penalties are assigned to misclassifying default cases using parameters such as **class_weight** or **scale_pos_weight**, encouraging the model to pay greater attention to the minority class during training.

Class counts before and after each resampling strategy (training split only, 176,400 rows):

Strategy	Non-default (0)	Default (1)
Before resampling	154,703	21,697
After SMOTE	154,703	154,703

Strategy	Non-default (0)	Default (1)
After SMOTETomek	150,868	150,868
After ADASYN	154,703	152,099
After random undersampling	21,697	21,697

scale_pos_weight for the boosting models (XGBoost, LightGBM, CatBoost) was set to 7.13, matching the actual imbalance ratio in the training data.

Model Selection

To identify the most suitable model for predicting loan default, several machine learning algorithms were evaluated using the same training and validation datasets. The models were selected to represent different learning approaches, allowing their performance to be compared under various class imbalance handling techniques.

- **Logistic Regression** was used as a baseline model because it is widely applied in credit risk prediction and produces easily interpretable results. It provides a benchmark for comparing the performance of more advanced models.

- **Decision Tree** was selected because it can capture nonlinear relationships between borrower characteristics, making it suitable for identifying patterns associated with loan default while remaining easy to interpret.

- **Random Forest** was included to improve prediction stability by combining multiple decision trees. This approach generally reduces overfitting and performs well on structured financial data.

- **XGBoost, LightGBM, and CatBoost** were chosen because boosting algorithms are known to achieve strong performance on tabular datasets. These models can learn complex relationships between borrower features and were expected to provide higher predictive accuracy than simpler models. LightGBM also offers faster training on large datasets, while CatBoost is particularly effective when working with categorical variables.

Comparing these models across different resampling strategies made it possible to identify the approach that most effectively detected potential defaulters while maintaining reliable overall performance.

Initial validation results (default 0.5 threshold)

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression (SMOTE)	0.607	0.170	0.566	0.262	0.626
Decision Tree (SMOTE)	0.549	0.181	0.758	0.293	0.701
Random Forest (SMOTE)	0.726	0.283	0.804	0.419	0.827
Random Forest (SMOTETomek)	0.730	0.287	0.805	0.423	0.830
Logistic Regression (Undersample)	0.609	0.171	0.567	0.263	0.626

Model	Accuracy	Precision	Recall	F1	AUC
Random Forest (Undersample)	0.809	0.369	0.782	0.502	0.870
Random Forest (class_weight)	0.838	0.412	0.755	0.533	0.866
XGBoost (scale_pos_weight)	0.838	0.416	0.788	0.545	0.871
LightGBM (scale_pos_weight)	0.874	0.491	0.795	0.607	0.894
CatBoost (scale_pos_weight)	0.735	0.276	0.710	0.397	0.798

The initial validation results show clear differences in performance across the ten model and imbalance-handling combinations. Overall, tree-based ensemble models consistently outperformed both the linear model and the single Decision Tree.

Among all models, **LightGBM with `scale_pos_weight`** achieved the best overall performance, with an **AUC of 0.894**, **87.4% accuracy**, and the highest **F1-score (0.607)**. This indicates that it was the most effective at distinguishing between default and non-default borrowers while maintaining a good balance between precision and recall. **XGBoost** produced similar results, achieving an AUC of **0.871**, making it another strong candidate for loan default prediction.

The Random Forest models also performed well, particularly when using **class weighting** or **random undersampling**, both of which outperformed the oversampling approaches. This suggests that adjusting the learning process through class weights was generally more effective than creating synthetic observations for this dataset.

In contrast, **Logistic Regression** produced the weakest results, with an AUC of approximately **0.63** regardless of the imbalance-handling method used. This is consistent with the earlier exploratory analysis, which suggested that the relationship between borrower characteristics and loan default is not purely linear. As a result, the more flexible tree-based models were better able to capture these complex patterns and provide more accurate predictions.

Based on these validation results, **LightGBM** was selected as the best-performing model for further optimization and threshold tuning.

Cross-validation check

Model	CV AUC (mean)	Std. dev.
Logistic Regression (SMOTE)	0.626	± 0.002
Decision Tree (SMOTE)	0.699	± 0.006
Random Forest (SMOTE)	0.820	± 0.003
Random Forest (Undersample)	0.864	± 0.002
Random Forest (class_weight)	0.864	± 0.001
XGBoost (scale_pos_weight)	0.870	± 0.002
LightGBM (scale_pos_weight)	0.891	± 0.003

To ensure that the model rankings were not influenced by a single train-validation split, a **5-fold stratified cross-validation** was performed using the **AUC** metric.
The results closely matched the initial validation scores. **LightGBM** remained the best-performing model with a mean **AUC of 0.891**, followed by **XGBoost (0.870)** and **Random Forest (0.864)**. In contrast, **Logistic Regression** consistently produced the lowest AUC.
The small standard deviations across all models (≤ 0.006) indicate stable performance across different folds, confirming that the results are reliable. Based on these findings, **LightGBM** was selected for further optimization.

2f. Model Evaluation and Optimization

Confusion matrices and per-class metrics

Model	Class	Precision	Recall	F1
Random Forest (class_weight)	0 (no default)	0.96	0.85	0.90
	1 (default)	0.41	0.75	0.53
XGBoost (scale_pos_weight)	0 (no default)	0.97	0.84	0.90
	1 (default)	0.42	0.79	0.54
LightGBM (scale_pos_weight)	0 (no default)	0.97	0.88	0.92
	1 (default)	0.49	0.80	0.61

Because only **12.3%** of the observations represent loan defaults, **accuracy alone is not a reliable measure of model performance**. For example, predicting every applicant as a non-defaulter would still achieve an accuracy of **87.7%**, while failing to identify any actual defaulters. For this reason, the evaluation focused on metrics that provide a more complete assessment of the model's ability to identify risky borrowers.

- **Precision** measures how many borrowers predicted as defaulters actually defaulted, indicating the reliability of positive predictions.
- **Recall** measures how many actual defaulters were correctly identified, making it particularly important since missing a high-risk borrower can result in financial losses.
- **F1-score** combines precision and recall into a single measure, providing a balanced evaluation when both false positives and false negatives matter.
- **AUC (Area Under the ROC Curve)** evaluates how well the model distinguishes between defaulters and non-defaulters across different classification thresholds.
- **H-measure** complements AUC by incorporating the different costs of misclassification, making it especially suitable for imbalanced classification problems such as loan default prediction.

ROC curves

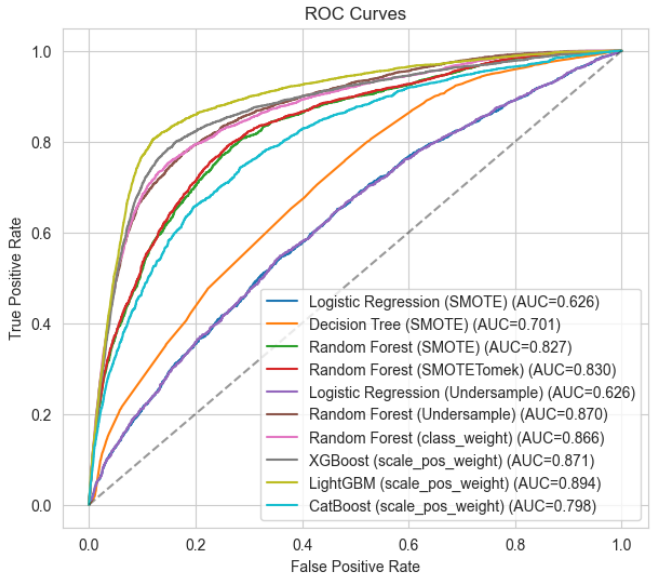


Figure 8. ROC curves for all ten validation-stage models.

Figure 8 presents the ROC curve, which illustrates how well the model distinguishes between borrowers who will default and those who will not across different classification thresholds. A curve that is closer to the upper-left corner indicates better classification performance, while a larger **AUC** reflects a stronger ability to separate the two classes.

Among the evaluated models, **LightGBM** achieved the highest **AUC of 0.894**, with its ROC curve remaining closest to the upper-left corner. This indicates that it consistently provides the best discrimination between default and non-default borrowers and confirms its suitability as the final prediction model.

Lift curve

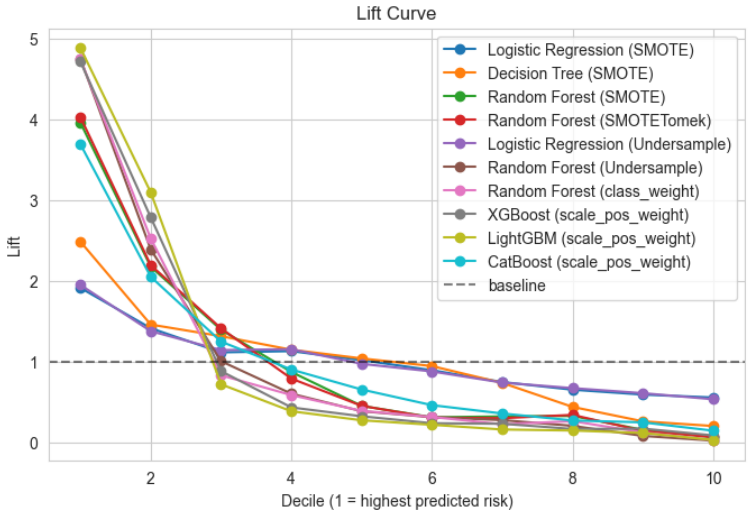


Figure 9. Lift by predicted-risk decile - how many more defaulters each model finds in its riskiest 10% of applicants, compared to random selection.

Figure 9 shows the lift curve, which evaluates how effectively the model ranks borrowers according to their likelihood of default. A higher lift in the first few deciles indicates that the model successfully places the highest-risk applicants at the top of the ranking.
The **LightGBM** model achieved the highest lift in the first decile, meaning that the 10% of applicants identified as highest risk contained nearly five times more defaulters than would be expected from random selection. This demonstrates the model's practical value for lenders, as it enables credit analysts to prioritize manual reviews on a small group of high-risk applicants while identifying a large proportion of potential defaults.

H-measure

To complement the AUC, **H-measure** was also used to evaluate model performance. While AUC measures how well a model separates defaulters from non-defaulters, it does not explicitly account for the different costs of classification errors. In loan default prediction, failing to identify a borrower who will default is generally more costly than incorrectly classifying a low-risk borrower as high risk. H-measure addresses this by evaluating all models under the same misclassification cost assumptions, making the comparison more consistent and appropriate for imbalanced datasets.

Model	H-measure
LightGBM (scale_pos_weight)	0.321
XGBoost (scale_pos_weight)	0.279
Random Forest (class_weight)	0.271
Random Forest (Undersample)	0.267
Random Forest (SMOTETomek)	0.185
Random Forest (SMOTE)	0.180
CatBoost (scale_pos_weight)	0.147
Decision Tree (SMOTE)	0.045
Logistic Regression (Undersample)	0.020
Logistic Regression (SMOTE)	0.019

LightGBM achieved the highest **H-measure (0.321)**, followed by **XGBoost (0.279)** and **Random Forest with class weighting (0.271)**. The ranking was consistent with both the **AUC** and **cross-validation AUC** results, providing further confidence that **LightGBM** is the most reliable model for this study.

Hyperparameter tuning

After comparing all candidate models, **LightGBM with `scale_pos_weight`** was selected for further optimization because it consistently achieved the highest performance in terms of **H-measure**, **AUC**, and **5-fold cross-validation AUC**. The agreement between these independent evaluation metrics suggested that its performance was stable rather than the result of a favourable train-validation split.

To further improve the model, **RandomizedSearchCV** was applied. This method efficiently searches different combinations of hyperparameters without testing every possible combination, making it well suited for large datasets while reducing computational time.

The search identified the following optimal parameters:

- **n_estimators = 400**
- **num_leaves = 127**
- **learning_rate = 0.1**
- **subsample = 1.0**
- **colsample_bytree = 0.6**

Using these parameters increased the cross-validated **AUC to 0.913**. On the validation set, the tuned model achieved an **AUC of 0.916**, **accuracy of 88.7%**, and an **F1-score of 0.630**, improving upon the default LightGBM model.
Since the dataset is imbalanced, **Macro F1** and **Weighted F1** were also considered. **Macro F1** evaluates both classes equally, ensuring that the minority default class has the same importance as the majority class. **Weighted F1** accounts for the class distribution, providing an overall performance measure while still considering class imbalance. Together, these metrics offer a more reliable assessment than accuracy alone.

Probability calibration

Although the tuned LightGBM model ranked borrowers effectively, its predicted probabilities were not perfectly calibrated because class weighting focuses on improving classification rather than producing accurate probability estimates.
To address this, **CalibratedClassifierCV** with **isotonic regression** was applied. Calibration adjusts the predicted probabilities so they better reflect the actual likelihood of default, making them more suitable for practical decision-making.
After calibration, the model achieved an **AUC of 0.923**, while **accuracy increased from 87.4% to 90.0%** and **Weighted F1 improved from 0.886 to 0.899**. More importantly, the calibration curve moved closer to the ideal diagonal line, indicating that predicted probabilities more closely matched the observed default rates. This provides greater confidence when using probability scores to assess borrower risk.

	Accuracy	F1 (macro)	F1 (weighted)	AUC
Raw (tuned) LightGBM	0.874	0.766	0.886	0.894
Calibrated LightGBM	0.900	0.766	0.899	0.923

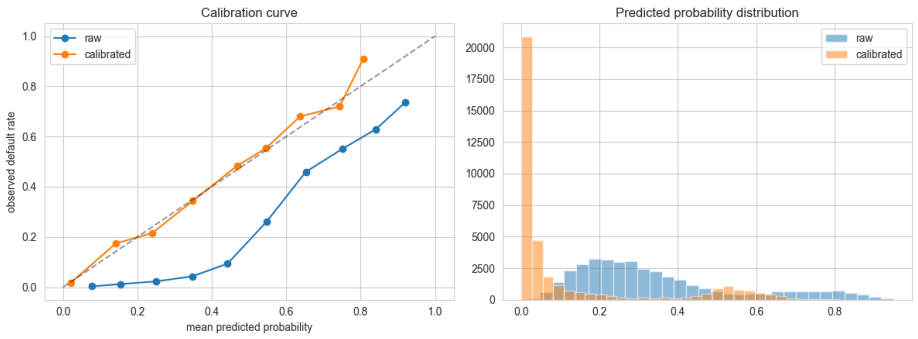


Figure 10. Calibration curve (left) and predicted-probability distributions (right), raw vs. calibrated.

Threshold tuning

Threshold rule	Cutoff	Accuracy	F1	Recall	Precision
Default	0.50	0.900	0.589	0.585	0.593
F1-optimal	0.33	0.896	0.641	0.755	0.557
Accuracy-optimal	0.47	0.901	0.623	0.667	0.585
Closest to 90% accuracy	0.49	0.900	0.604	0.621	0.589

Machine learning models classify observations by comparing predicted probabilities with a decision threshold. While the default threshold is **0.50**, this is not always the best choice for an imbalanced dataset where identifying defaulters is the main objective.

Several threshold values were evaluated to determine the best balance between **precision** and **recall**. The **F1-optimal threshold of 0.33** produced the strongest overall performance, increasing the **F1-score to 0.641** and **recall to 0.755**. This means the model identified substantially more borrowers who eventually defaulted while only sacrificing a small amount of overall accuracy.

Although a higher threshold slightly increased accuracy, it also reduced recall by missing more high-risk borrowers. Since the primary objective of this study is to detect potential loan defaults, the **0.33 threshold** provides a more appropriate balance for credit risk assessment.

Stacking ensemble

Model	Accuracy	F1	F1 (macro)	F1 (weighted)	AUC
Stacked Ensemble	0.896	0.623	0.781	0.901	0.903

As a final experiment, a **stacking ensemble** combining **LightGBM, XGBoost, CatBoost, and Random Forest** was developed, with **Logistic Regression** acting as the meta-learner. The purpose was to investigate whether combining several strong models could improve predictive performance beyond the tuned LightGBM model.

The stacked model achieved an **AUC of 0.903**, **accuracy of 89.6%**, and an **F1-score of 0.623**. Although these results were strong, they did not outperform the **calibrated LightGBM model**, which achieved the highest **AUC (0.923)** and better overall classification performance.

Considering the additional computational cost and increased implementation complexity of maintaining multiple models, the stacked ensemble offered no practical advantage. Therefore, the **calibrated and tuned LightGBM model** was selected as the final model because it provided the best balance between predictive performance, reliability, and simplicity.

Final model comparison

Model	Accuracy	Precision	Recall	F1	F1 (macro)	F1 (weighted)	AUC	H-measure
Calibrated LightGBM	0.90	0.59	0.58	0.59	0.77	0.90	0.92	0.35
Tuned LightGBM	0.89	0.53	0.78	0.63	0.78	0.90	0.92	0.34
Stacked Ensemble	0.90	0.56	0.69	0.62	0.78	0.90	0.90	0.33
LightGBM (scale_pos_weight)	0.87	0.49	0.80	0.61	0.77	0.89	0.89	0.32
XGBoost (scale_pos_weight)	0.84	0.42	0.79	0.54	0.72	0.86	0.87	0.28
Random Forest (class_weight)	0.84	0.41	0.75	0.53	0.72	0.86	0.87	0.27
Random Forest (Undersample)	0.81	0.37	0.78	0.50	0.69	0.83	0.87	0.27
Random Forest (SMOTETomek)	0.73	0.29	0.81	0.42	0.62	0.77	0.83	0.19
Random Forest (SMOTE)	0.73	0.28	0.80	0.42	0.62	0.77	0.83	0.18
CatBoost (scale_pos_weight)	0.73	0.28	0.71	0.40	0.61	0.78	0.80	0.15
Decision Tree (SMOTE)	0.55	0.18	0.76	0.29	0.48	0.62	0.70	0.04
Logistic Regression (Undersample)	0.61	0.17	0.57	0.26	0.50	0.68	0.63	0.02
Logistic Regression (SMOTE)	0.61	0.17	0.57	0.26	0.50	0.67	0.63	0.02

This table compares the performance of all evaluated models. Overall, the **Calibrated LightGBM** achieved the best results, with the highest **H-measure (0.35)**, **AUC (0.92)**, and **90% accuracy**, making it the most reliable model for loan default prediction. Although the **Tuned LightGBM** achieved slightly higher recall, the calibrated model produced more reliable probability estimates, making it better suited for practical use. The **Stacked Ensemble** also delivered strong performance but did not outperform the calibrated LightGBM, while requiring greater computational complexity. It should be noted that the precision and recall values reported for the calibrated model are based on the **default threshold of 0.50**. In practice, the final deployed model uses the **optimized threshold of 0.33**, which increases recall to **0.755** and enables the model to identify a greater proportion of borrowers at risk of default.

Feature importance

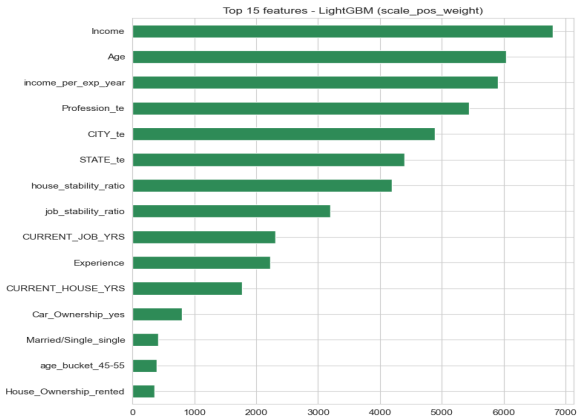


Figure 11. Top 15 features by importance, tuned LightGBM model.

Figure 11 shows the 15 most important features used by the final **Tuned LightGBM** model. Income and the engineered financial stability features were the strongest predictors of loan default. The target-encoded variables for **city**, **state**, and **profession** also ranked highly, indicating that location and occupation contain important patterns related to default risk. These findings demonstrate the ability of LightGBM to capture complex, non-linear relationships that were not evident from the earlier correlation analysis.

2g. Results and Recommendations

Final test set evaluation

The final model was evaluated on the test set, which remained completely unseen during model development. This provides an unbiased assessment of how the model is expected to perform on new loan applicants.

Metric	Value
Model deployed	Calibrated LightGBM (scale_pos_weight, tuned)
Decision threshold	0.33
Accuracy	0.894
Precision (default class)	0.549
Recall (default class)	0.770
F1 (default class)	0.641
AUC	0.921

The **Calibrated LightGBM** model, using the optimized **0.33 decision threshold**, achieved an **accuracy of 89.4%**, **precision of 0.549**, **recall of 0.770**, **F1-score of 0.641**, and an **AUC of 0.921**. These results are very similar to those obtained on the validation set, indicating that the model generalizes well to unseen data and has not been overfitted during training or optimization. The consistently high recall also confirms that the model is effective at identifying borrowers who are most likely to default, making it suitable for supporting lending decisions.

Key risk factors identified

The analysis identified several borrower characteristics that are strongly associated with loan default. Among all features, the target-encoded variables for profession, city, and state were the most influential in the final LightGBM model, indicating that both occupation and geographical location play an important role in predicting default risk. This suggests that borrowers with similar financial profiles may still have different levels of risk depending on where they live and the type of work they perform.

Marital status also showed a noticeable relationship with loan default**.** Single applicants had a higher default rate (12.53%) than married applicants (10.25%)**,** suggesting that marital status may provide additional information about financial stability.

Although car ownership was found to be statistically associated with default risk, the effect size was relatively small. This indicates that, while the relationship is significant, car ownership alone is not a strong predictor and should be considered alongside other borrower characteristics.

The analysis also found that older applicants and those with longer work experience or job tenure were slightly less likely to default. However, these relationships were relatively weak, meaning they are more useful when combined with other variables than as standalone indicators.

Interestingly, income alone was not a statistically significant linear predictor of default**.** Instead, its importance emerged when combined with other features, such as employment experience and financial stability measures. This helps explain why the tree-based LightGBM model outperformed Logistic Regression, as it was able to capture these complex, non-linear relationships.

Recommendations for lenders

- The findings suggest that the **Calibrated LightGBM** model can be used as a decision-support tool to improve credit risk assessment. Rather than automatically rejecting loan applications, lenders should use the model's risk scores to **prioritize applications for manual review**. As shown by the lift curve, the model effectively concentrates high-risk borrowers within the highest-risk group, allowing underwriters to focus their efforts where they are most needed.

- The selected **0.33 decision threshold** provided the best balance between identifying potential defaulters and limiting false alarms. However, this threshold should not be viewed as fixed. It can be adjusted depending on a lender's risk tolerance and the relative cost of approving a borrower who later defaults versus incorrectly flagging a low-risk applicant.

- To maintain reliable performance, the model should also be **retrained and recalibrated periodically**, as borrower behavior, economic conditions, and regional risk patterns may change over time. Regular updates will help ensure that the predicted probabilities continue to reflect actual default risk.

- Finally, while occupation and geographical location were identified as important predictors, these variables should be used to support lending decisions rather than replace professional judgement. Their predictive value reflects historical patterns within the dataset and should not be interpreted as evidence that an individual borrower will default based solely on their profession or location. Additionally, when communicating model performance to stakeholders, the focus should remain on **recall, precision, AUC, and lift**, as these metrics provide a more meaningful assessment than accuracy for an imbalanced loan default dataset.